Animal Shelter Intake, Stay & Outcome Analysis

Interactive Power BI dashboard analyzing animal intake, shelter stay, and outcome patterns between 2018 and 2022.

Tools: Power BI · Power Query · DAX · M · CSV
Data Modeling: Relational Data Model
Data Visualization: Interactive dashboards, KPI cards, charts, slicers, and filters

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1. Project Overview

The goal of this project was to develop an interactive Power BI report analyzing the flow of animals through a shelter, from intake to outcome.

The report analyzes animal intake and outcome records from 2018 to 2022, focusing primarily on dogs and cats, which make up the majority of the shelter population.

The analysis explores animal characteristics, intake patterns, shelter population, length of stay, and adoption outcomes. The report aims to provide insights that can support informed decisions regarding animal care, shelter capacity, resource planning, and adoption activities.

2. Business Questions

The analysis focuses on the following questions:

Animal Intakes

- How many animals are admitted to the shelter, and which animal types are most common?
- What are the most common intake types and recorded health conditions?

- What are the most common breeds, colors, and age groups among admitted animals?
- How do intake characteristics differ between dogs and cats?

Intake Trends

- How does the number of animal intakes change over time?
- Are there periods when intake volume increases or decreases significantly?
- Are there seasonal patterns in animal intakes?
- How do intake trends differ between dogs and cats?

Shelter Stay

- How long do animals typically stay in the shelter?
- How many animals leave the shelter within the first week?
- Does length of stay differ by age group and animal type?
- Which age groups tend to stay in the shelter the longest?
- How does the shelter population change over time?

Outcomes and Adoptions

- How do animals leave the shelter?
- What is the average age of animals at outcome and adoption?
- Which breeds and colors are most common among adopted dogs and cats?
- Are outcome patterns different between dogs and cats?
- Are there seasonal patterns in animal outcomes?

3. Data

The project uses CSV files containing animal intake and outcome records provided as part of an analytical training course.

The report covers records from 2018 to 2022 and focuses primarily on dogs and cats, while also including other animal types in overall analyses.

The main tables used in the report include:

- Intakes — records and characteristics of animals entering the shelter.
- Outcomes — records of animals leaving the shelter and their outcomes.
- Date — a calendar table used for time-based analysis.

Additional helper tables were created to support the report.

4. Data Preparation and Modeling

4.1 Data was cleaned and transformed in Power Query.

Main steps included:

- importing CSV files,
- removing unnecessary columns,
- changing data types,
- creating additional columns containing animal age information,
- merging tables to calculate Length of Stay,
- creating a calendar table,
- creating helper tables,
- preparing the data for analysis and visualization.

4.2 Data Model

The report uses a relational data model connecting animal intake and outcome records with a calendar table.

The model supports analysis of animal flow over time, as well as comparisons by animal type, age group, breed, and other characteristics.

4.3 DAX Measures

DAX measures were created to analyze:

- total intakes and outcomes,
- adoptions, transfers, and euthanasias,
- average age at intake and outcome,
- average length of stay,
- length of stay by age group and animal type,
- shelter population over time.

These measures are used in KPI cards and interactive visualizations throughout the report.

4.4 Visualizations and Interactivity

The report includes interactive visualizations designed to explore animal intake, shelter stay, and outcome patterns. The visualizations include: KPI cards, line charts, bar charts, donut charts, year and month filters, interactive visual filtering.

Users can explore trends over time and compare animal characteristics, including differences between dogs and cats. Selecting categories in interactive visuals also allows users to examine related data across the report.

5. Report Structure

1 — Intakes Overview

Provides an overview of animals entering the shelter, including total intake volume and trends over time.

The page includes:

- total intakes and intake trends,
- average age at intake,
- animal type distribution,
- intake type distribution,
- intake condition distribution,
- sex distribution,
- most common breeds and colors among dogs and cats.

The page helps identify the main characteristics of animals entering the shelter and patterns in intake volume.

2 — Shelter Stay

Analyzes the shelter population and the length of time animals remain in care, with a focus on comparing dogs and cats.

The page includes:

- animals by length of stay,
- average length of stay by age group,
- intakes by age group,
- sex distribution at intake for dogs and cats,
- average time to adoption for dogs and cats,
- most common breeds and colors among dogs and cats,
- average age at adoption,
- shelter population by age group over time.

The page supports comparisons of shelter stay patterns between dogs and cats and helps identify age groups associated with longer stays.

3 — Outcomes Overview

Focuses Provides an overview of how animals leave the shelter and how outcomes change over time.

The page includes:

- total outcomes and outcome trends,
- average age at outcome,

- outcome type distribution,
- animal type distribution,
- sex distribution.

The page helps explore the main outcome types and compare outcome patterns across animal groups.

6. Key Insights

The analysis identified several notable patterns in animal intake, shelter stay, and outcomes.

Intake Volume and Animal Characteristics:

- During the five-year period, the shelter recorded more than 130,000 animal intakes, including over 70,000 dogs and over 53,000 cats.
- The average age at intake was approximately 1 year and 11 months.
- Cats were admitted at an average age of approximately 1 year and 3 months, while dogs were admitted at approximately 2 years and 6 months.
- The shelter recorded more dog intakes than cat intakes during the analyzed period. Dogs were also older on average at intake.
- The vast majority of animals entering the shelter were recorded as healthy.
- The most common intake types were Stray and Owner Surrender.

Breed and Color Patterns:

- The most common cat breed and color combination at intake was Domestic Shorthair Mix with Brown Tabby coloring. This was also the most common combination among adopted cats.
- Among dogs, Pit Bull Mix with Black/White coloring was the most common breed and color combination at intake.
- The most frequently adopted dog breed was Labrador Retriever Mix, differing from the most common dog breed and color combination at intake.

Intake Trends and Seasonality:

- The highest monthly intake volume occurred in May 2019, with 1,935 animals.
- Intake volume then declined sharply, reaching its lowest point in April 2020, with 290 animals.
- From April 2020 onward, intake volume generally showed a gradual upward trend, although it did not return to the 2019 peak.
- Intake patterns showed seasonality, with the highest volumes generally occurring in the second quarter, particularly in May.
- Intake volume tended to decrease during the third quarter and was lowest during the winter months, especially December, January, and February.

Length of Stay:

- The average length of stay was approximately 37 days for cats and 26 days for dogs.
- Despite these averages, most animals left the shelter within the first week.
- The most frequently admitted cat age group was 0–6 months. Young cats had one of the longest average stays, followed by senior cats aged 7+ years.
- The most frequently admitted dog age group was 1–3 years.
- Senior dogs aged 7+ years had the longest average length of stay among dogs.

Outcomes and Adoptions:

- The average age at adoption was approximately 1 year and 1 month for cats and 1 year and 10 months for dogs.
- Cats most commonly left the shelter through adoption or transfer.
- Dogs were adopted approximately twice as often as they were transferred or returned to their owners.
- Among other animal types, euthanasia was the most common recorded outcome.
- Outcome trends broadly followed intake trends, with higher activity during warmer months and lower activity during winter.
- Seasonal patterns were more pronounced and regular among cats than among dogs.

7. Analytical Value

The report provides an interactive overview of animal flow through the shelter and highlights differences in intake, shelter stay, and outcomes, particularly between dogs and cats.

The analysis can help:

- identify changes in intake volume and animal characteristics,
- understand seasonal patterns in intakes and outcomes,
- compare shelter stay patterns across animal types and age groups,
- identify breeds and characteristics common among admitted and adopted animals,
- understand how animals leave the shelter and at what age.

These insights can support informed decisions regarding animal care, shelter capacity, resource planning, and adoption activities.